Mathematical Primer

For "Prediction Markets Are Fisher Markets"

Five ideas that make the paper work

What this document covers. The paper uses five mathematical ideas. Each builds on the previous. If you understand these five things, you understand the entire paper — everything else is either an application of these ideas or context for why they matter.

#	**Idea**	**Unlocks**
1	Complementary slackness	Theorems 3, 4, 8
2	Fenchel conjugates	§2.2, §2.3, §2.8 (Theorems 1, 2, 7)
3	KKT conditions	§2.4, §5.2 (Theorems 3, 8)
4	The budget absorption trick	Theorem 8 part (3) — the punchline
5	Why log makes it convex	The entire §3 → §5 arc

1. Minting Cost and the LP: Why $\max_k D_k$?

Before the five ideas, one piece of setup from the paper (§2.1) needs unpacking: the clearing LP and its minting cost.

1.1. What Minting Is

A prediction market has K mutually exclusive outcomes. Minting creates one share of *every* outcome at cost \$1. This is fairly priced: exactly one outcome resolves to \$1 at settlement, so a complete set is always worth \$1.

The key constraint: you cannot create shares of a single outcome. Every mint produces one share of each.

1.2. What D_k Actually Is

D_k looks like "demand for outcome k" but it's more precise than that. It is the *net* creation requirement — how many new shares of outcome k the exchange must create, after netting buys against sells:

$$D_k = \sum_{i \in \text{ buy}(k)} q_i - \sum_{j \in \text{ sell}(k)} q_j$$

Some shares come from sellers (they already hold shares and are giving them up). D_k is what's left over — the shortfall that sellers can't cover.

> **Example.** Outcome A has 100 shares of buy orders filled and 40 shares of sell orders filled. Net: $D_A = 100 - 40 = 60$. The exchange must *create* 60 new A-shares. The other 40 came from sellers.

Crucially, **the optimizer controls $\boldsymbol{D_k}$** through its choice of fills $\boldsymbol{q}$. It can make D_k small by filling fewer buy orders or filling more sell orders. D_k is not external demand — it's a consequence of the optimizer's decisions.

If $D_k \leq 0$ (more selling than buying), no new shares of outcome k are needed — shares are being returned/destroyed.

1.3. Why $M \geq D_k$

M is the number of complete sets minted. Each mint produces one share of *every* outcome. After M mints, you have M new shares of each outcome available.

D_k is how many new shares of outcome k the fills require (as computed above).

The constraint $M \geq D_k$ says: **new shares produced ≥ new shares needed**, for each outcome. You must mint enough complete sets to cover the largest shortfall. You don't need $M \geq D_k$ for outcomes where sellers already cover the demand — only where the net is positive.

1.4. The LP Is a Tradeoff

The clearing LP is:

$$\max_{\boldsymbol{q}, M} \quad \underbrace{\sum_i w_i q_i}_{\text{welfare from fills}} \quad - \quad \underbrace{M}_{\text{minting cost}} \qquad \text{s.t.} \quad D_k(\boldsymbol{q}) \leq M \ \forall k, \quad \boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$$

The optimizer chooses *both* which orders to fill *and* how much to mint. It's a cost–benefit tradeoff:

- **Filling orders creates welfare** ($w_i q_i > 0$ for profitable orders).
- **Filling orders creates net demand** (D_k increases), which requires minting.
- **Minting costs money** ($-M$ in the objective).
- **The optimizer is free to fill nothing.** Setting $\boldsymbol{q} = \boldsymbol{0}$ gives $D_k = 0$, $M = 0$, objective $= 0$. This is always feasible.

The optimizer fills orders only when the welfare gained exceeds the minting cost incurred. The whole LP is the optimizer asking: **which fills create enough welfare to justify the minting they require?**

> **Example.** Suppose filling a buy order creates \$0.60 of welfare but requires minting that costs \$0.80 (because no sellers offset it). The optimizer skips this order — it destroys value. But if a sell order offsets half the minting need, the net cost drops to \$0.40, and the fill becomes worthwhile.

1.5. Why $M = \max_k D_k$ at the Optimum

The constraint *allows* minting more than needed ($M \geq D_k$, not $M = D_k$). But M is subtracted from the objective — it's a cost. The optimizer pushes M down as far as possible, which means $M = \max_k D_k$ at the optimum (the tightest the constraints allow). The $\geq$ is the LP formulation; the optimizer tightens it to equality.

This is also complementary slackness at work: the dual variable of $D_k \leq M$ is the price p_k, and $p_k > 0$ only where $D_k = M$ (the binding constraints).

1.6. Why Is the Surplus Free?

When you mint 70 sets but only need 30 B-shares, the remaining 40 B-shares are surplus. Why don't they cost anything? Because complete sets cost \$1 and pay out \$1 — the surplus shares of losing outcomes are worthless at settlement, and the one winning outcome's surplus was already paid for by the mint. The $\max_k D_k$ accounting captures this exactly: you pay for the most-demanded outcome, and everything else comes free with the complete sets.

2. Complementary Slackness

You have a constraint — say, a fill can't exceed its maximum quantity:

$$q_i \leq \overline{Q}_i$$

The optimizer assigns a *dual variable* (also called a multiplier) μ_i^+ to this constraint. Think of it as the "price" of the constraint: how much better would the objective be if you relaxed this bound by one unit?

Complementary slackness says:

Rule. At the optimum, at least one of these is zero: the multiplier, or the slack.

$$\mu_i^+ \cdot \left(\overline{Q}_i - q_i\right) = 0$$

Two cases:

Situation	Slack	Multiplier
Constraint is tight ($q_i = \overline{Q}_i$)	Zero — no room left	Can be positive — relaxing would help
Constraint is loose ($q_i < \overline{Q}_i$)	Positive — room left	Must be zero — relaxing is pointless

Intuition. "Relaxing" means raising the ceiling — changing $\overline{Q}_i$ from 100 to 101, giving the constraint more room. If you're not using the full capacity, a higher ceiling wouldn't help — so the "price" of more capacity is zero. If you *are* at full capacity, a higher ceiling might actually help, so the price can be positive.

Where the paper uses this:
- **Theorem 4** (self-financing): the dual variable of the minting constraint $D_k \leq M$ is the price p_k. Complementary slackness says $p_k > 0$ only where $D_k = M$ — price is positive only for the highest-demand outcome.
- **§2.4** (clearing rule): gives the three cases (fully filled / unfilled / marginal).
- **Theorem 8** (budget absorption): the $\lambda_i^+ q_i$ terms that make spending $\leq$ budget.

3. Fenchel Conjugates

This is the engine of §2. A Fenchel conjugate transforms a function from "quantity space" to "price space."

3.1. The Definition

Given a function $f(x)$, its conjugate is:

$$f^*(p) = \sup_x [p \cdot x - f(x)]$$

You sweep over all x, computing "revenue minus cost," and take the best.

3.2. The Meaning

Think of $f(x)$ as a *cost function* — producing quantity x costs $f(x)$. Now someone offers price p per unit. Your profit from producing x is:

$$\text{profit} = \underbrace{p \cdot x}_{\text{revenue}} - \underbrace{f(x)}_{\text{cost}}$$

The conjugate $f^*(p)$ is your **maximum profit at price $\boldsymbol{p}$** — the best you can do by choosing the optimal production quantity.

> **Key idea.** f lives in quantity space. f^* lives in price space. They encode the same information from dual perspectives. For convex f, the conjugate of the conjugate gives back f.
>
> $$\begin{array}{ccc} f(x) & \xleftrightarrow{\text{conjugate}} & f^*(p) \\ \text{cost of producing } x & & \text{max profit at price } p \\ \text{quantity space} & & \text{price space} \end{array}$$

3.3. Worked Example: Minting Cost (Theorem 1)

Our cost function is $V(\boldsymbol{D}) = \max_k D_k$ — the minting cost. We want the conjugate:

$$V^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \left[\sum_k p_k D_k - \max_k D_k \right]$$

You're selling outcome shares at prices p_k and paying minting cost $\max_k D_k$. What demand vector $\boldsymbol{D}$ maximizes profit?

Case: $\boldsymbol{p}$ is a probability vector ($\sum p_k = 1$, all $p_k \geq 0$).

Revenue $= \sum p_k D_k$ is a weighted average of the D_k's with weights summing to 1. A weighted average can never exceed the maximum:

$$\sum p_k D_k \leq \max_k D_k$$

So profit ≤ 0 for every $\boldsymbol{D}$. At $\boldsymbol{D} = \boldsymbol{0}$, profit $= 0$. Best profit: $V^*(\boldsymbol{p}) = 0$.

Case: $\sum p_k > 1$.

Set all $D_k = t$, send $t \to \infty$. Revenue $= t \sum p_k$. Cost $= t$. Profit $= t(\sum p_k - 1) \to \infty$.

Arbitrage: you mint complete sets at \$1 each and sell the shares for $\sum p_k >$ \$1. Scale up for unbounded profit.

Case: $\sum p_k < 1$.

Set all $D_k = -t$ (buy back shares), send $t \to \infty$. Profit $= t(1 - \sum p_k) \to \infty$.

Reverse arbitrage: buy a complete set of shares for $\sum p_k <$ \$1 and redeem for \$1.

Case: some $p_k < 0$.

Drive $D_k \to -\infty$ for that outcome. Someone is paying you to take shares. Infinite profit.

Result:

$$V^*(\boldsymbol{p}) = \begin{cases} 0 & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

This is the simplex indicator. Its meaning: **the only prices where you can't arbitrage the minting mechanism are probabilities.** The probability axiom is a no-arbitrage condition, not a modeling choice.

3.4. Worked Example: LMSR Cost (Theorem 2)

Replace minting cost with LMSR: $C_b(\boldsymbol{D}) = b \ln \sum \exp(D_k/b)$. Same game:

$$C_b^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}} \left[\sum p_k D_k - b \ln \sum \exp(D_k/b)\right]$$

For the minting cost (Theorem 1), $V = \max_k D_k$ has a kink — it's not differentiable where two D_k's are tied — so we had to reason case-by-case. The LMSR cost C_b is smooth (that's the whole point of entropy smoothing), so we can use calculus: find the maximum by setting the derivative to zero.

The expression inside the sup — "linear in $\boldsymbol{D}$" minus "convex in $\boldsymbol{D}$" — is *concave* in $\boldsymbol{D}$. For a concave function, any point where the derivative is zero is the global maximum (not just a local one — this is what convexity/concavity buys you). So setting the derivative to zero finds *the* answer:

$$\frac{\partial}{\partial D_k}\left[\sum p_k D_k - C_{b(\boldsymbol{D})}\right] = p_k - \frac{\exp(D_k/b)}{\sum_j \exp\left(D_j/b\right)} = 0$$

$$\therefore \quad p_k = \frac{\exp(D_k/b)}{\sum_j \exp\left(D_j/b\right)}$$

This is the **softmax**. The conjugate asks "what demand vector maximizes profit at prices $\boldsymbol{p}$?" and the answer is: the one where prices equal the softmax of demand.

Invert ($D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp\left(D_j/b\right)$), substitute back, and after algebra:

$$C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k$$

Negative Shannon entropy. Finite only on the simplex (softmax always outputs probabilities). Most negative at uniform $p_k = 1/K$ (maximum entropy = most uncertain prices).

3.5. The Duality Picture

Cost (quantity space)		**Conjugate (price space)**	**Character**
$V = \max_k D_k$	$\longleftrightarrow$	$V^* = \delta_\Delta$	Hard wall: probabilities or nothing
$C_b = b \ln \sum \exp(D_k/b)$	$\longleftrightarrow$	$C_b^* = b \sum p_k \ln p_k$	Soft penalty: non-uniform costs more

As $b \to 0$: the soft penalty hardens into the wall.

Why this matters for the paper. In §2.8, the clearing problem is rewritten in price space as:

$$\min_{\boldsymbol{p} \in \Delta}\left[W^*(\boldsymbol{p}) + C_b^*(\boldsymbol{p})\right]$$

The entropy term C_b^* is *strictly convex* — it curves. That curvature forces a unique minimizer. This is Theorem 7: without budgets, prices are always unique. The entire §3 obstacle is about budgets breaking this.

4. KKT Conditions

You want to maximize $f(x)$ subject to $x \in [0, \overline{Q}]$. Calculus says "set $f'(x) = 0$" — but what if the max is at the boundary?

KKT handles boundary optima with multipliers:

$$f'(x) = \mu^+ - \mu^-$$

where $\mu^+, \mu^- \geq 0$ with complementary slackness ($\mu^+ (x - \overline{Q}) = 0$, $\mu^- x = 0$).

Case	**Condition**	**Meaning**
Interior ($0 < x < \overline{Q}$)	$f'(x) = 0$	Just calculus — both multipliers zero
At ceiling ($x = \overline{Q}$)	$f'(x) \geq 0$	Function still rising — you'd want more but you're capped
At floor ($x = 0$)	$f'(x) \leq 0$	Function falling — you want none

In the paper (§2.4). The objective per order is $w_i q_i - C_b(\boldsymbol{D})$. For buy order i on outcome k:

$$\frac{\partial}{\partial q_i}[w_i q_i - C_b] = L_i - \underbrace{\frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)}}_{p_k}$$

The derivative is just $L_i - p_k$ (limit price minus clearing price). KKT says:

Case	**Derivative**	**In English**
Fully filled ($q_i = \overline{Q}_i$)	$L_i - p_k \geq 0$	Limit above price — fill everything
Unfilled ($q_i = 0$)	$L_i - p_k \leq 0$	Limit below price — fill nothing
Marginal ($0 < q_i < \overline{Q}_i$)	$L_i - p_k = 0$	Limit equals price — this order sets the price

This is the Uniform Clearing Price rule: buy if your limit exceeds the price.

Why this matters. Theorem 8 uses the exact same KKT structure, but with B_k/U_k replacing w_i. If you understand the above, you understand the Theorem 8 proof — it's the same three cases with one extra chain-rule step.

5. The Budget Absorption Trick

This is the punchline of the paper (Theorem 8, part 3). We replace linear welfare $\sum w_i q_i$ with $B_k \ln U_k$ where $U_k = \sum_{i \in \mathrm{MM}_k} L_i q_i$ is MM k's total weighted fill. An important modeling choice: MM_k contains only MM k's *buy* orders (with $L_i > 0$). Sell orders from MMs are treated as retail (linear welfare). This ensures $U_k > 0$ and aligns with the Fisher market analogy, where agents are consumers (buyers of goods).

5.1. Step 1: KKT for the Log Objective

For MM buy order i belonging to MM k, differentiate $B_k \ln U_k$ with respect to q_i:

$$\frac{\partial}{\partial q_i}[B_k \ln U_k] = B_k \cdot \frac{1}{U_k} \cdot L_i$$

This is the chain rule: derivative of ln is $1/U_k$, derivative of U_k w.r.t. q_i is L_i, times the weight B_k.

The full KKT also includes the minting cost derivative. Since this is a buy order, it increases $D_{m(i)}$ by q_i, costing $p_{m(i)}$ per share. With box multipliers:

$$\frac{B_k L_i}{U_k} - p_{m(i)} = \lambda_i^+ - \lambda_i^-$$

5.2. Step 2: The Telescoping

Multiply both sides by q_i:

$$\frac{B_k L_i q_i}{U_k} - p_{m(i)} q_i = \lambda_i^+ q_i - \lambda_i^- q_i$$

Note: $\lambda_i^- q_i = 0$ by complementary slackness (if $q_i > 0$ then $\lambda_i^- = 0$; if $\lambda_i^- > 0$ then $q_i = 0$). So the last term vanishes. Now sum over all $i \in \mathrm{MM}_k$:

$$\frac{B_k}{U_k} \cdot \underbrace{\sum_{i \in \mathrm{MM}_k} L_i q_i}_{\text{this is } U_k \text{ by definition}} - \underbrace{\sum_{i \in \mathrm{MM}_k} p_{m(i)} q_i}_{\text{capital spent on purchases}} = \underbrace{\sum_{i \in \mathrm{MM}_k} \lambda_i^+ q_i}_{\geq 0}$$

The left side: U_k cancels, leaving $B_k - \sum p_{m(i)} q_i$.

$$B_k - \sum_{i \in \mathrm{MM}_k} p_{m(i)} q_i = \sum_{i \in \mathrm{MM}_k} \lambda_i^+ q_i \geq 0$$

$$\therefore \quad \sum_{i \in \mathrm{MM}_k} p_{m(i)} q_i \leq B_k$$

Each MM's capital deployed on purchases is **at most** B_k, with equality when no fill hits its upper bound.

That's it. Five lines of algebra. The budget constraint was never imposed — it *emerged* from the first-order conditions of the log objective.

5.3. Why This Works: Intuition

$\ln(U_k)$ has a singularity at $U_k = 0$: the slope goes to ∞.

Near $U_k = 0$:	Marginal value B_k/U_k is huge — fill desperately
As U_k grows:	Marginal value B_k/U_k shrinks — diminishing returns
At optimum:	Marginal value per dollar = price for every active order

Because of the B_k scaling, this balance point is reached exactly when total spending = B_k. Bigger budget → more fill before saturation. The budget isn't a constraint; it's a *consequence of the curvature.*

6. Why Log Makes It Convex

Now we connect everything. The risk-neutral model has a non-convexity problem (§3). The log model doesn't (§5). Here's why.

6.1. The Risk-Neutral Problem

Objective: $\sum w_i q_i - C_b(\boldsymbol{D})$.

Component	Curvature	Source
$\sum w_i q_i$ (linear welfare)	None — flat	No help
$-C_b$ (minting cost)	Concave, strength $O(1/b)$	Entropy smoothing
Budget constraint $c(p) \cdot q \leq B_k$	Non-convex, strength $O(1/b)$	Price × quantity

The only curvature comes from $-C_b$, which scales as $O(1/b)$. But the budget non-convexity *also* scales as $O(1/b)$ — they grow at the same rate. Neither dominates. **Deadlock.**

At $b = 0$ (LP), there is zero curvature and the budget constraint is fully non-convex. No hope.

6.2. The Risk-Averse Fix

Objective: $\sum B_k \ln U_k + \text{retail} - C_b(\boldsymbol{D})$.

Component	Curvature	Source
$\sum B_k \ln U_k$ (log welfare)	Strictly concave, $O(B_k/U_k^2)$	ln — independent of b
$-C_b$ (minting cost)	Concave, strength $O(1/b)$	Entropy smoothing — bonus
Budget constraint	**Gone**	Absorbed by §4 trick

Two things change simultaneously:

1. **The budget constraint disappears** from the feasible set. The bilinear $c(p) \cdot q \leq B_k$ is gone — absorbed into the objective (§4 trick). No more non-convex feasible set.
2. **Strict curvature appears** in the objective. The ln provides $O(B_k/U_k^2)$ concavity, independent of b. This guarantees a unique maximum.

The $-C_b$ term is now pure bonus — extra concavity on top of what ln already provides. Even at $b = 0$ (no entropy smoothing at all), the program is still strictly concave:

$$\max_{q \in \mathcal{C}} \quad \sum_k B_k \ln U_k + \sum_{j \notin \text{ MM}} w_j q_j - \max_k D_k$$

The $-\max_k D_k$ is concave (not strictly), but ln carries the strict concavity alone. **No annealing needed.**

6.3. The Full Picture

	Risk-neutral	**Risk-averse**
Welfare	$\sum w_i q_i$ (linear)	$\sum B_k \ln U_k$ (log)
Budget constraint	Explicit, bilinear, non-convex	Absent — absorbed into objective
Curvature source	$-C_b$ only ($O(1/b)$)	ln ($O(B_k/U_k^2)$, b-independent)

At $b = 0$	Zero curvature, non-convex — hopeless	Strict curvature, convex — fine
Unique prices?	Not always (Prop. 2)	Always (Thm. 8)
Complexity	GNEP (intractable in general)	Convex program (polynomial)

One modeling change — linear welfare → log welfare — removes the non-convexity and adds strict curvature. The paper's thesis: this isn't a trick. The linear model was the fiction.

This primer covers the five mathematical ideas in "Prediction Markets Are Fisher Markets." For the full proofs, formal statements, and economic motivation, see the paper itself.